LSTM vs Transformer Encoder for AG News Text Classification

Term Project

Introduction to Deep Learning

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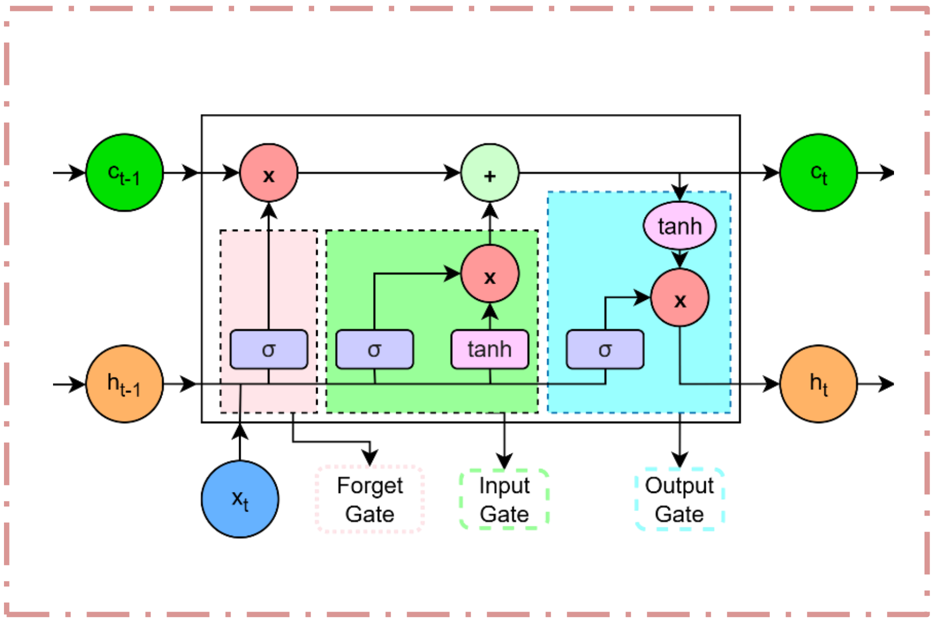
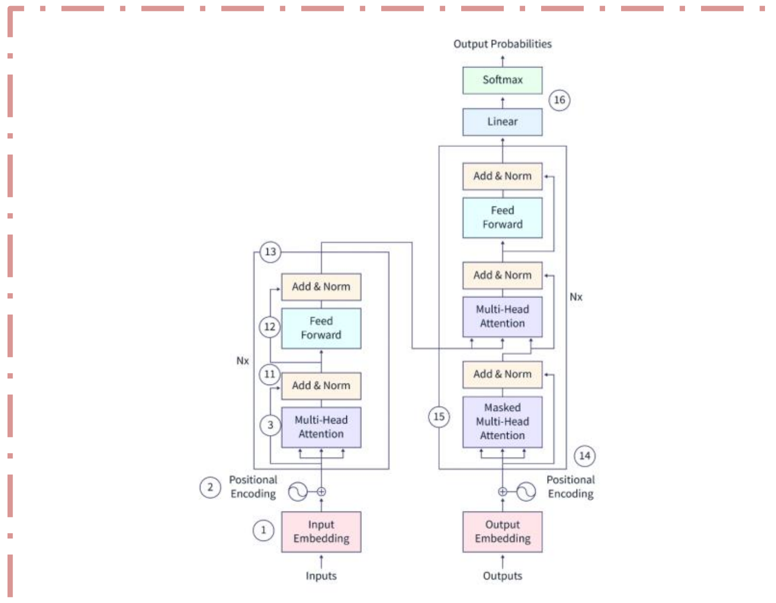
Why compare LSTMs and Transformer Encoder?

- Both models are common sequence classifiers, but they process text differently.
- The LSTM carries information recurrently through hidden states.
- The Transformer Encoder uses self-attention to connect tokens directly.
- The goal is to compare behavior, not only final accuracy.



LSTM



Transformer Encoder

Dataset

- AG News is a news topic classification dataset.
- Each example is a short text item, and the label is one of four classes:
  - World
  - Sports
  - Business
  - Sci/Tech
- We sourced the dataset from Hugging Face exclusively.

Split	Source	Samples	Use
Train	90% of official training split	108,000	Model fitting and train-only vocabulary construction
Validation	10% of official training split	12,000	Model selection, ablation comparison, and interpretation
Test	Official test split	7,600	Final evaluation only

Preprocessing

Both models receive text after the same preprocessing pipeline:

- Raw text is lowercased and tokenized with a regular expression tokenizer.
- A vocabulary is built from training text only.
- The vocabulary is capped at 20,000 entries.
- Two reserved tokens are used:
  - *<pad>* at index 0.
  - *<unk>* at index 1.
- Tokens are converted to integer IDs.
- Sequences are truncated or padded to a fixed maximum length.
- An attention mask marks real tokens as 1 and padding positions as 0.

Model Development

- LSTM: embedding layer, one-layer bidirectional LSTM, masked mean pooling, dropout, classifier.
- Transformer Encoder: embeddings, learned positional encoding, two encoder layers, four heads, classifier.
- Trainable parameters: LSTM 2.83M; Transformer Encoder 2.97M.
- Both models are trained from scratch.

2.83M
LSTM parameters

2.97M
Transformer parameters

- Important Hyperparameters:

Adam
Optimizer

$\mathbf{1 \times 10^{-3}}$
Learning Rate

6
Epochs

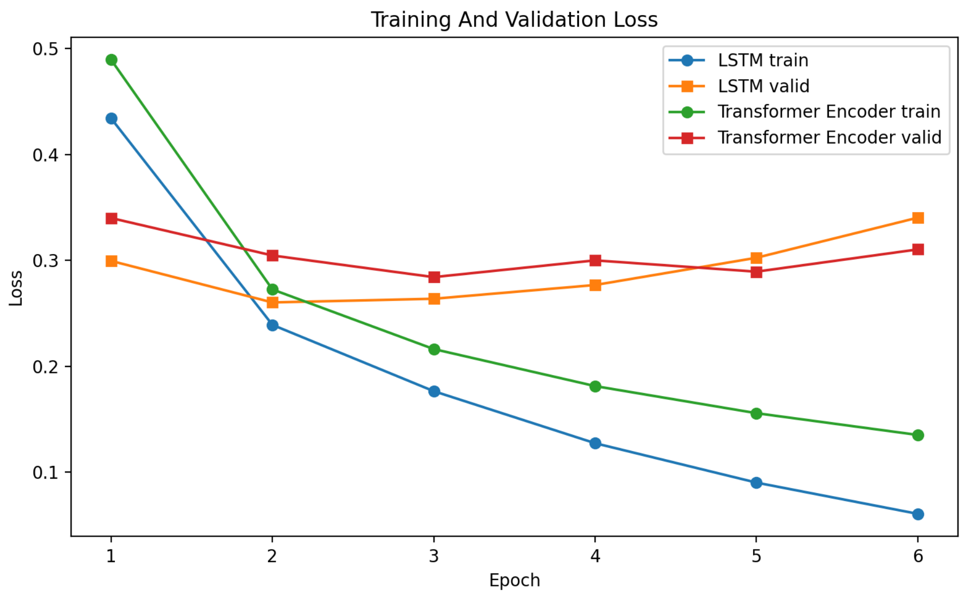
CrossEntropyLoss
Loss Function

Main Results

The Transformer Encoder has the best final test metrics, but the margin is small:

- Accuracy lead: 0.0005, or 0.05 percentage points.
- Macro F1 lead: about 0.0008.

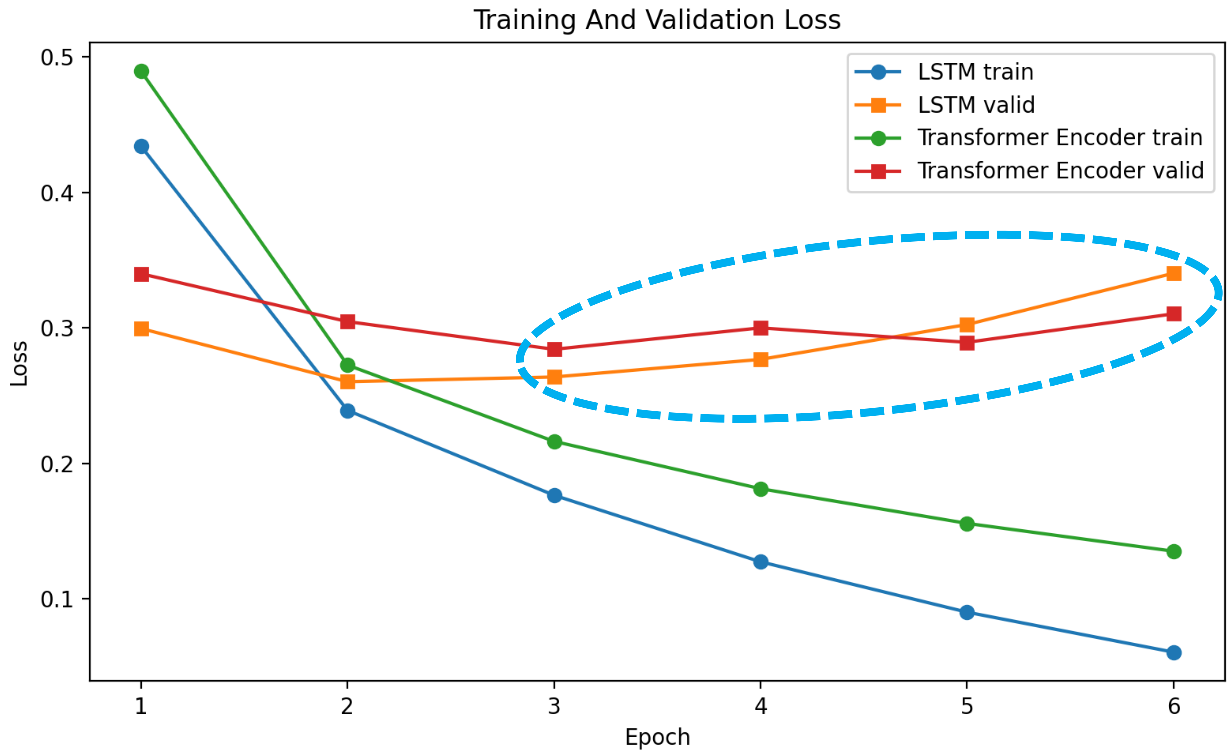
Both models perform very similarly. The Transformer Encoder is slightly ahead in this run, but the result is not a decisive architectural win.



Model	Split	Loss	Accuracy	Macro F1
LSTM	Validation	0.3402	0.9113	0.9108
Transformer Encoder	Validation	0.3103	0.9118	0.9115
LSTM	Test	0.3459	0.9120	0.9118
Transformer Encoder	Test	0.3066	0.9125	0.9126

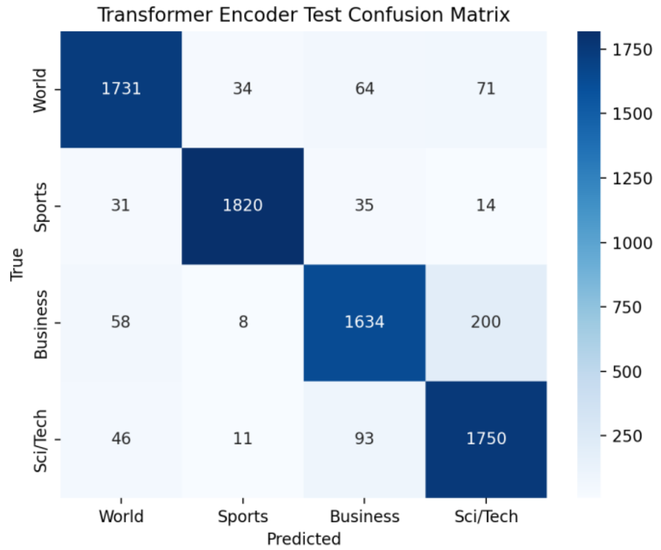
Main Results

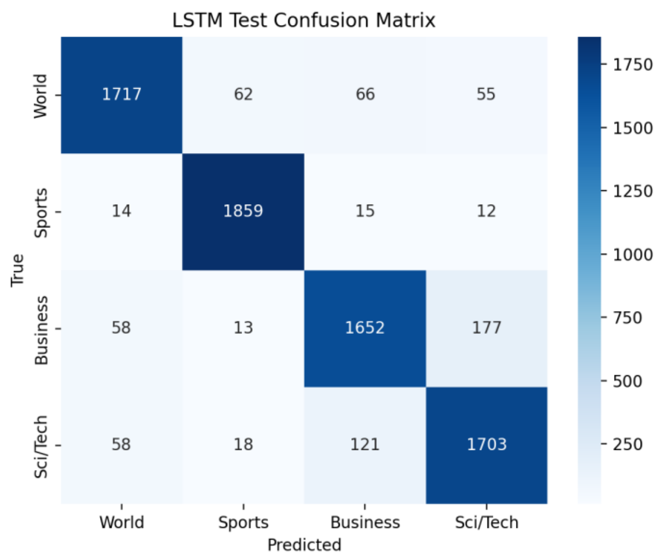
- Training loss continues to decrease across 6 epochs.
- Validation loss rises late in training.
- This suggests early stopping or stronger regularization as the logical next improvement.



Main Results

- Sports is the easiest class for both models.
- Business and Sci/Tech overlap because many company stories include technology terms.
    - LSTM: 177 Business examples predicted as Sci/Tech, and 121 Sci/Tech examples predicted as Business.
    - Transformer Encoder: 200 Business examples predicted as Sci/Tech, and 93 Sci/Tech examples predicted as Business.
- The pattern is consistent with the selected failure examples.





Ablation Studies

Two Ablation Studies were run:

- Variable Training Set Fraction – 25%, 50%, 100%
- Variable Sequence Length – 128, 256

Training Set Fraction

Model	Train fraction	Accuracy	Macro F1
LSTM	25%	0.8719	0.8713
Transformer Encoder	25%	0.8699	0.8696
LSTM	50%	0.8944	0.8940
Transformer Encoder	50%	0.8901	0.8895
LSTM	100%	0.9089	0.9082
Transformer Encoder	100%	0.9146	0.9142

- 25% data, LSTM is slightly better.
- 50% data, LSTM is still better.
- 100% data, Transformer Encoder is better.

LSTM is more stable in lower-data settings, while the Transformer Encoder benefits more from the full dataset.
However, the differences are modest, so the result should be described as a trend rather than universal proof from this result.

Sequence Length

Model	Max length	Loss	Accuracy	Macro F1
LSTM	128	0.3402	0.9113	0.9108
Transformer Encoder	128	0.3103	0.9118	0.9115
LSTM	256	0.3535	0.9065	0.9058
Transformer Encoder	256	0.3304	0.9062	0.9059

- No improvement going from 128 to 256. Both models performed worse at 256 tokens.

Likely, AG News texts are short enough that 128 tokens already capture the useful signal.

Longer sequences increase computation and may introduce noisy or weakly useful tokens.

Failure Analysis

#	Input excerpt	True	LSTM	Transformer	Likely reason
1	Live: Olympics day four Richard Faulds and Stephen Parry are going fo...	World	Sports	Sports	Sports terms dominate the short text even though the dataset section is World.
2	Intel to delay product aimed for high-definition TVs SAN FRANCISCO --...	Business	Sci/Tech	Sci/Tech	A technology company story overlaps Business and Sci/Tech vocabulary.
3	Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn w...	Sci/Tech	World	Sci/Tech	Weather prediction language mixes science terms with world-event framing.
4	Some People Not Eligible to Get in on Google IPO Google has billed it...	Sci/Tech	Business	Sci/Tech	Finance and technology cues conflict; the IPO framing points toward Business.
5	Teenage T. rex's monster growth Tyrannosaurus rex achieved its massiv...	Sci/Tech	Sci/Tech	Business	Very short science item gives limited context; the Transformer overweighs business-like wording.
6	IBM to hire even more new workers By the end of the year, the computi...	Sci/Tech	Sci/Tech	Business	Company hiring story overlaps labor/business cues and computing-company Sci/Tech label.

- Surface cues can dominate the true news section. For example, an Olympics-related text labeled World was predicted as Sports by both models.
- Business and Sci/Tech overlap heavily when a story discusses a technology company, IPO, chip delay, software company, or computing-related business event.
- Short science or technology items can be ambiguous because there is less context for the model to average over.

Conclusion

- Both models reach about 91.2% final test accuracy.
- The Transformer Encoder is slightly better on final test accuracy and macro F1.
- The LSTM is stronger at 25% and 50% training data.
- The Transformer Encoder benefits more at 100% training data.
- Increasing max sequence length from 128 to 256 hurts validation performance in this run.
- The main error pattern is Business vs Sci/Tech ambiguity.
- Both models show overfitting by the final epoch.
- Future work: early stopping, regularization, and class-level error reduction.


Thank you!
Thanks!
Thank you so much!
Thanks a lot!